

Nikita Kazeev

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Research Scientist at the intersection of AI and Physics

WORK EXPERIENCE

PERFOMAX	AI Advisor	2026–present
CONSTRUCTOR GROUP	Machine Learning Consultant	2021–present
NATIONAL UNIVERSITY OF SINGAPORE	Postdoc under Kostya Novoselov	2022–present

Leadership

- Program Chair for the [AI4X 2025/6 conferences](#)
- Main organizer of the [ICLR 2025 Workshop on Machine Learning Multiscale Processes](#)

Wyckoff Transformer, a generative model for materials [ICML 2025](#) · [GitHub](#)

- Developed a coordinate-free permutation-invariant autoregressive Transformer encoder model for material generation and property prediction based on symmetry inductive bias
- Increased materials discovery yield (S.U.N.) by 1.24x
- Accelerated inference by 4 orders of magnitude, to 0.05 GPU ms per structure
- Implemented the model in PyTorch; production-ready package published on [PyPI](#)
- Orchestrated large-scale (10k structures) DFT computations with VASP at an HPC cluster
- Led a team of 6

ML for defects in 2D crystals [npj 2D Mater. Appl.](#) · [npj Comput. Mater.](#) · [patent](#) · [GitHub](#)

- Conceived a novel sparse representation of crystals with defects
- Achieved a 3.7x energy prediction error reduction
- Optimized training and inference: 4x less memory, 8x fewer GPU operations
- Developed code for reproducible parallel running of machine learning experiments
- Led a team of 3

⟨CERN | YANDEX → HSE UNIVERSITY⟩ *Researcher* 2014–2022
[2025 Breakthrough Prize in Fundamental Physics](#) as a member of LHCb.

Generative models uncertainty estimation [paper](#) · [conference](#) · [code 1](#) · [code 2](#)

- Developed the first methods for estimating uncertainty of conditional GANs.
- Led a team of 3 students.

Generative models for fast simulation [paper 1](#) · [paper 2](#) · [paper 3](#) · [code](#)

- Developed a GAN-based machine learning model for high-fidelity simulation of a Cherenkov detector trained on tabular data.
- Achieved approximately 10^5 speed-up compared to the ab-initio simulator.

Machine learning on noisy data [paper 1](#) · [paper 2](#) · [code](#)

- Developed a rigorous way to train ML algorithms on data with the label-noise model common in high-energy physics.

Muon identification at the LHCb experiment at CERN [paper](#) · [poster](#) · [IDAO-2019](#)

- Solved a classification problem over tabular data with noisy labels under timing constraints.
- Developed a gradient boosting model that reduced the false positive rate by 30% in the critical low-track-momentum region.
- Integrated the model into LHCb production, mostly in C++.
- Packaged the work as a data science competition problem for IDAO-2019.

CatBoost aka fighting biases with dynamic boosting [paper](#)

- Set up a distributed system for running experiments for the team with Bash and Python.
- Studied gradient boosting improvements with experiments on toy data.

EDUCATION

HSE University & Sapienza Università di Roma <i>PhD in Computer Science and Physics (Double Degree)</i> Supervisors: Andrey Ustyuzhanin & Barbara Sciascia Thesis: Machine Learning for particle identification in the LHCb detector	2016–2020
Product Management course at Yandex <i>Focused coursework in product strategy and execution.</i>	2018
Yandex School of Data Analysis <i>Master's-level CS programme</i> Algorithms, machine learning, deep learning, and distributed systems.	2013–2015
Moscow Institute of Physics and Technology <i>MS in Physics</i> Optimisation of data processing of the LHCb experiment.	2014–2016
<i>BS in Physics</i> Wave-packet molecular dynamics of electrons in nonideal plasma.	2010–2014
International Junior Science Olympiad (IJSO) Silver medal	Korea, 2008

MENTORSHIP

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- Mentored 9 students ([1](#), [2](#), [3](#), [4](#), [5](#), [6](#), [7](#), [8](#), [9](#)), 3 interns, and student workshop projects ([1](#), [2](#)).
 - Student council member from 2013 to 2016, leading several dormitory-scale IT projects.
 - Co-PI of a \$3.4m AI Singapore grant on multiscale machine learning.

TEACHING & OUTREACH

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- Pitch research to the general public through major newspaper coverage ([1](#), [2](#)), blog posts ([1](#), [2](#)), a [science slam](#), interviews ([1](#), [2](#)), and a public [talk](#).
 - Taught at the Machine Learning in High Energy Physics Summer School in [2015](#), [2016](#), [2017](#), [2018](#), [2019](#), [2020](#), [2021](#), and [2023](#).
 - Lecturer in the [award-winning](#) Coursera Advanced Machine Learning [specialisation](#)
 - Taught and developed an introductory data analysis [course](#) at HSE.
 - Delivered more than 20 [conference talks](#).

SERVICE

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- Reviewer for NeurIPS, RSC Advances, Machine Learning: Science and Technology, Digital Discovery
 - Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

SKILLS

- **Generative modeling for scientific data** — GANs, autoregressive transformers, and diffusion; from fast detector simulation to symmetry-aware crystal generation.
- **Uncertainty quantification** — built the first methods for estimating the uncertainty of conditional GANs; calibrated generative and discriminative models.
- **ML on structured, non-text data** — tabular, graph, and atomistic representations, spanning gradient boosting to transformers.
- **Inference and systems at scale** — C++ production integration, large-scale DFT/VASP on HPC clusters, and reproducible experiment pipelines.
- **Physics domain depth** — high-energy and condensed-matter physics; first-principles intuition that shapes model design.
- **Research leadership** — lead multidisciplinary teams, chair conferences, and mentor students from idea to publication.

ML & AI • Python • PyTorch • C++ • HPC • Physics • Research Writing & Speaking • Leadership & Mentorship

PRESENTING AT ICML 2026

July 11, 12:15–13:30 — Nikita Kazeev and Ian Babich. [“Foresight-Phys: A Benchmark for Forecasting the Results of Physical Experiments”](#). In: *Forecasting as a New Frontier of Intelligence Workshop at ICML 2026*.

July 11, 15:00–16:30 — Nikita Kazeev and Phan Bui Nhat Huyen. [“Position: Align AI to Our Aspirations, Not Our Flaws”](#). In: *Pluralistic Alignment Workshop at ICML 2026*.

July 11, 15:30–17:00 — Nikita Kazeev and Andrey Ustyuzhanin. [“The Geometry of Reasoning Failure: Predicting Agent Errors from Semantic Scale Trajectories”](#). In: *Statistical Frameworks for Uncertainty in Agentic Systems Workshop at ICML 2026*.

July 11, 10:50–11:50 & 15:55–16:55 — Artem Dembitskiy*, Shuya Yamazaki*, Artem Maevskiy*, Nikita Kazeev*, Roman A. Eremin, Semen Budennyi, Kedar Hippalgaonkar, Antonio Helio Castro Neto, and Andrey E. Ustyuzhanin. [“Generative Pipeline for Discovering Solid-State Battery Materials with Universal Atomistic Potentials”](#). In: *AI for Science Workshop at ICML 2026*. *Equal Contribution.

July 11, 11:40–13:30 & 16:10–17:00 — Nikita Kazeev and Andrey Ustyuzhanin. [“Uncertainty Quantification for LLM Agents via Semantic Abstraction Trajectories”](#). In: *Structured Probabilistic Inference & Generative Modeling Workshop at ICML 2026*.

SELECTED PUBLICATIONS

Nikita Kazeev, Wei Nong, Ignat Romanov, Ruiming Zhu, Andrey E Ustyuzhanin, Shuya Yamazaki, and Kedar Hippalgaonkar. [“Wyckoff Transformer: Generation of Symmetric Crystals”](#). In: *Proceedings of the 42nd International Conference on Machine Learning*. Ed. by Aarti Singh, Maryam Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff, and Jerry Zhu. Vol. 267. Proceedings of Machine Learning Research. PMLR, 13–19 Jul 2025, pp. 29495–29526.

Nikita Kazeev, A. R. Al-Maeni, I. Romanov, et al. [“Sparse representation for machine learning the properties of defects in 2D materials”](#). In: *npj Computational Materials* 9 (2023), p. 113.

Maxim Borisyak and Nikita Kazeev. [“Machine Learning on data with sPlot background subtraction”](#). In: *Journal of Instrumentation* 14.08 (2019). Alphabetic order, I’m the corresponding author.

Denis Derkach, Nikita Kazeev, Fedor Ratnikov, Andrey Ustyuzhanin, and Alexandra Volokhova. [“Cherenkov detectors fast simulation using neural networks”](#). In: *Nuclear Instruments and Methods in Physics Research Section A* (2019). Alphabetic order, I’m the corresponding author.